Dr. Shahadat Hussain

Additive Manufacturing · Materials Engineering · Smart Manufacturing

📍 Abu Dhabi, UAE ✉️ contact@shahadathussain.com 🌐 shahadathussain.com 🔗 linkedin.com/in/shahadathussain
ORCID 0000-0002-4355-2169 | ISNI 0000-0005-3020-7229 | GND 1397994843 | Scopus 56380929800 | ResearcherId I-3091-2017

PROFESSIONAL SUMMARY

Advanced Manufacturing & Materials Engineer with 6+ years of experience in additive manufacturing (LPBF, FFF, SLA), materials characterization, and mechanical testing. Specializes in NiTi shape memory alloys, TPMS metamaterial structures, and composite AM — with hands-on expertise across 6+ industrial-grade 3D printing systems and a full suite of characterization and testing equipment. Combines deep materials science knowledge with practical data science skills (Python, MATLAB, ML) to support Industry 4.0 and digital manufacturing initiatives. Currently based in Abu Dhabi, UAE with deep familiarity with the region's industrial and research landscape through years of work at leading UAE institutions. Seeking to transition into an industrial engineering or R&D role where advanced materials and manufacturing technologies create tangible operational impact.

CORE COMPETENCIES

Additive Manufacturing (LPBF · FFF · SLA)	**Materials Characterization (SEM · XRD · DSC) & Mechanical Testing**	**Industry 4.0 / Smart Manufacturing**
NiTi & Cu Shape Memory Alloys	**TPMS / Metamaterial Structures**	**CAD & Process Parameter Optimization**
Data Science & Machine Learning	**Interpenetrating Phase Composite Manufacturing**	**Lab Management & Safety (EHS)**

PROFESSIONAL EXPERIENCE

Research Associate — Additive Manufacturing & Advanced Materials
Sep 2024 – Present

New York University Abu Dhabi (NYUAD) · Abu Dhabi, UAE

- Design, fabricate, and characterize large-scale carbon fiber–reinforced PLA/PPA composite metamaterial structures (gyroid & primitive TPMS lattices) using FFF — targeting aerospace, automotive, and structural applications.
- Conduct quasi-static compression, flexural, modal, and frequency-response testing to evaluate strength, energy absorption, and vibration behavior of printed composite components.
- Operate and maintain 10+ industrial 3D printing systems including Raise3D Pro3 Plus HS, Stratasys J850/F370, Formlabs Form3, and Bambu Lab Carbon X1.

Postdoctoral Research Fellow — Metal 3D Printing & Smart Materials
Feb 2020 – Jun 2024

Khalifa University · Abu Dhabi, UAE

- Fabricated NiTi shape memory alloy TPMS lattice structures (Primitive & Gyroid) via Laser Powder Bed Fusion (EOS M400); systematically optimized laser power, scan speed, and energy density to control microstructure and phase transformation.
- Performed full materials characterization pipeline on 50+ printed samples: JEOL 7610F SEM, FEI Quanta 3D FIB-SEM (SEI/BSE/EDS), Bruker D2 & Rigaku MiniFlex XRD, Mettler Toledo & Setaram DSC, Leica optical microscopy.
- Executed static (Instron 5969, 50 kN), fatigue (Instron 8872, 25 kN), and compression (MTS 100 kN) testing campaigns; developed printing quality maps for AM NiTi components.
- Published 4 peer-reviewed journal papers (Materials & Design, Journal of Manufacturing Processes, Materials, JMEP); 450+ citations, h-index 10. Reviewed 30+ manuscripts across 15 leading journals.
- Completed comprehensive EHS Lab Safety training: chemical handling, HF acid etching, cryogenic substances, risk assessment, and emergency response.

Project Fellow — Smart & Functional Materials
Jun 2013 – Aug 2014

CSIR-Advanced Materials & Processes Research Institute (AMPRI) · Bhopal, India

- Synthesized Cu-Al-Mn/Cu-Al-Ni shape memory alloys via vacuum induction melting as part of a Government of India (12th Five Year Plan) funded project; performed hot/cold rolling, heat treatment, quenching, and metallographic preparation.
- Characterized phase transformations and mechanical properties using SEM, XRD, DSC, and hardness/tensile testing; contributed to 5 SCI journal publications and established research collaboration with IIT Madras.

Engineering Intern — Steel Plant Operations
Jun – Jul 2011

Bokaro Steel Plant, Steel Authority of India Limited (SAIL) · Jharkhand, India

- Gained direct industrial exposure to large-scale integrated steel manufacturing processes, plant operations, and quality systems.

EDUCATION

PhD — Materials Science & Technology
2014 – 2019

AcSIR / CSIR-AMPRI, Bhopal, India · *Thesis: Cu-based shape memory alloys — composition, processing, microstructure & shape memory properties*

BEng — Mechanical Engineering
2008 – 2012

UIT-RGPV, Bhopal, India

TECHNICAL SKILLS

Additive Manufacturing Systems
- EOS M400 (LPBF/SLM — metal powder bed)
- Stratasys F370 / J850 (FDM / PolyJet)
- Raise3D Pro3 Plus HS / Forge1 (FFF)
- Formlabs Form3 (SLA) · Ultimaker S1

Materials Characterization
- JEOL 7610F SEM · FEI Nova NanoSEM
- FEI Quanta 3D FIB-SEM (SEI / BSE / EDS)
- Bruker D2 & Rigaku MiniFlex XRD
- Mettler Toledo DSC1 · Setaram DSC
- Leica Optical Microscope · Bruker Q4 Tasman

Mechanical Testing & Processing
- Instron 5969 static (50 kN) · 8872 fatigue (25 kN)
- MTS 100 kN UTM · Controls Simplex 350 & Automax
- Vacuum induction melting · Two-roll rolling
- Heat treatment · Metallography · Etching

Design, Simulation & Computing
- SolidWorks · FreeCAD
- Materialise Magics (build preparation) · GrabCAD Print
- Bambu Studio · PreForm · IdeaMaker · Cura
- Python (Pandas, NumPy, Scikit-learn)
- MATLAB · SQL · Linux · Shell scripting · LaTeX

CERTIFICATIONS

- Additive Manufacturing Specialization — Arizona State University (WSLFXJ6FELF9)
- IBM Data Science Professional Certificate (4R8YJ5YWQ0NH)
- Responsible Conduct of Research — CITI Program, Florida
- Google AI Essentials Certificate (L7UXWNCY24BU)
- Python for Everybody Specialization — University of Michigan (K3LMQEN5VEKX)
- Lab Safety (EHS) — Khalifa University (2021)

SELECTED PUBLICATIONS [500+ CITATIONS · H-INDEX: 10 · ORCID: 0000-0002-4355-2169]

1. **Hussain S, Mehraj H, Karathanasopoulos N, Moustafa MA.** *Do chopped fibers matter? Process-Structure-Property Relationships and Thermal Annealing Effects in Fiber-Reinforced TPMS Lattices.* Progress in Additive Manufacturing, 2026.

2. **Hussain S, Alagha AN, Zaki W.** *Phase Transformation Behavior of NiTi TPMS Lattices Fabricated by LPBF.* J. Materials Engineering & Performance, 2024.

3. **Hussain S, Alagha AN, Haidemenopoulos GN, Zaki W.** *Microstructural & Surface Analysis of NiTi TPMS Lattice Sections Fabricated by LPBF.* J. Manufacturing Processes, 102:375–386, 2023.

Full list: scholar.google.com/citations?user=tQNSWaAAAAAJ · *Peer reviewer for 15 journals incl. Additive Manufacturing, Materials & Design, Scripta Materialia.*

LANGUAGES & ADDITIONAL INFORMATION

Languages: English (Professional) · Hindi (Native) · Urdu (Native)
Visa Status: UAE Residence Visa holder — available to interview immediately.
GitHub: github.com/drshahadathussain · **ResearchGate:** researchgate.net/profile/Shahadat-Hussain